

Exploring the Milky Way

Astronomy Activity

TEACHER NOTES

Level: Moderate to advanced level for High School.

Objective: The Radio Horn Telescope is designed to measure neutral hydrogen in the Milky Way Galaxy. If you truly wish to explore the full DSPIRA Radio Horn Telescope experience all the way through to using it as a scientific instrument, then this would be the right science sequence to go through. This sequence will build a base knowledge of the Milky Way and move onto activities using the Radio Horn Telescope.

Recommended Sequence of Activities

DSPIRA Activity	General Astronomy Students And Physics Students	Goal
A1	How Fast Are We Moving?	This activity was in the Earth's Motion sequence, but it did have the students determine the speed of the Sun around the Milky Way
A2	Celestial Coordinate Systems	If the students are heading towards making observations from their location, a basic lesson of coordinate systems is needed. Galactic Coordinates are covered here.
E3	Using Stellarium to Find the Galactic Center	This also is in the Earth Motion Sequence, but it is additional support for understanding observing the Galaxy
MW1	Introduction to the Milky Way - What can we see?	Just a basic chance for the students to get basic information about the Milky Way off the internet. There is also a first pass at understanding the Galactic Coordinate system used for studying the Milky Way.
MW2	Introduction to the Milky Way - What can't we see?	This is mainly about modeling the Galaxy's motion with respect to the Sun's reference frame.

DSPIRA Activity	General Astronomy Students And Physics Students	Goal
MW3	A Model of the Motion within the Milky Way	This is a paper model that allows the students to determine relative radial speeds of stars at locations around the Milky Way Galaxy. A collaborative class effort can produce a model very similar to current research in Milky Way Astronomy.
MW4	Creating a Galactic Plane Survey using the Radio Horn	With some planning, you and your students can get a complete dataset through the Galactic Plane (visible from your location) in a few periods of class.

Pre-Knowledge of Students:

- 1) Basic Algebra and calculator skills.
- 2) Ruler and Protractor skills
- 3) Graph interpretation
- 4) Doppler Effect - The topic is not taken on in this sequence. It is assumed done in previous module

Key Terms:

Galactic Equator, Galactic Longitude, Sagittarius A*, Galactic Quadrants, Meridian, Altitude/Azimuth

Related Videos for Teacher:

[Activity E3- Quick tutorial for Stellarium - Web Version](#)

[Activity E3- Stellarium - Web or Download version?](#)

[Activity MW2 - Computer Simulation of Galaxy Rotation](#)

[Activity MW3 - Modeling the Motion of the Milky Way - Part 1](#)

[Activity MW3 - Modeling the Motion of the Milky Way - Part 2](#)

[Activity MW3 - Modeling the Motion of the Milky Way - Part 3](#)

Teacher Preparation:

This sequence is not to be taken lightly. There has to be some devotion to the goal of understanding the Milky Way, its motions, and current scientific understanding and observational data. The modeling

activities were an idea brewed up during the Summer DSPIRA workshop and much effort was put into creating visual aids and modeling exercises to bring some tough-to-grapple perspectives and heavy mathematics down to a palatable level for students to show the connection to modeling and observation.

If your school has created a DSPIRA Radio Horn Telescope (or more), this would be a great chance for extra credit work for students to take actual Galactic Data. There is a lot of Science Fair potential in these activities for students truly interested in science beyond their grades.

Activity A1- How Fast are We Moving?

- This is a stand alone activity with it's own ACT.A1 [Teacher Notes Page](#) (click)

Activity A2- Celestial Coordinate Systems

- Another foundation activity that works in a few DSPIRA Sequences ACT.A2 [Teacher Notes Page](#)

Activity E3- Using Stellarium to Find the Galactic Center

- This activity was covered in the Earth Around the Sun Sequence. The following is from those notes:
- Activity E3 - This activity is meant to be doable from just the instructions and access to Stellarium Web online version.
- Activity E3 - If you are venturing deep into DSPIRA activities and eventually working with the Radio Horn Telescope, you may wish to take on working with the downloadable version (still free). It would be good to have it downloaded on a few laptop computers that might also be used with the Horn Telescope.

Activity MW1- Introduction to the Milky Way - Part 1

- This could be a stand alone activity if you just want to introduce basic statistics of the MWG

Interlude - Doppler Effect -

- There is much in DSPIRA that is Doppler specific, but a general lesson has not been created yet. They are quite common, and easy to create. One suggested [Doppler Video](#)

Activity MW2- Introduction to the Milky Way - Part 2

- This activity is centered around understanding relative motion within the Galaxy through viewing a video capture of a computer simulation.
- If this Milky Way lesson sequence is the first one you are doing, then you will want to talk about the Doppler Effect.
- Again, the students should be able to get through the activity with minimal assistance.
- See video up in Related Videos

Activity MW3- Modeling Motion in the Milky Way

- There is a lot of prep work for this lab activity. For all pieces to fit together, the model needs to be printed to scale. The model was designed for each step along the orbit (dots on circle) to be 1 cm each. This brings the model to about 31 cm (12+ inches) across. So, you will either need to print it a a special printer, or ask it to be printed on about 4 regular sheets of paper that would need to be cut and taped together.
- Laminating would be a good idea. The pins avoid the necessity to draw on the model, but that doesn't always stop students. Lamination will help.
- Getting $\frac{1}{2}$ " or $\frac{3}{4}$ " foam board insulation, or a few layers of cardboard would be handy for this activity for pin placement.
- Three Part video will be vital for assistance. See Related Videos

Activity MW4 - Creating a Galactic Plane Survey using the Radio Horn

- There is no document at this point, but if you and your students have gotten to this point in the sequence, there is opportunity for student/teacher exploration...Not all science is a guided tour!